

THE UNIVERSITY OF ZAMBIA
MAT 1110 - Foundation Mathematics and Statistics
for Social Sciences
LECTURE NOTES - 2023
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Chapter 1

SETS

1.1 Introduction

In this chapter, we will review some basic concepts on set theory. After this chapter, a student is expected to define a set correctly, perform different set operations; intersection, union, complement and subsets, represent sets on Venn diagrams, shade sets as well as use the De Morgans laws to simplify set operations.

Definition 1.1.1. A set is a collection of well defined objects. The objects of a set are called elements or members of that set. Sets are commonly represented using set braces , $\{\}$.

Example 1.1.1. *The set containing the elements 1,2,3 and 4 is given by $\{1,2,3,4\}$.*

Note 1.1.1. In a set, the order in which elements are listed is not important.

Remark 1.1.1. To show that an element is a member of the set, we use the symbol $\in$. E.g in the preceding example, we can say 4 is in the set $\{1,2,3,4\}$ and so we write $4 \in \{1,2,3,4\}$. When an element is not in a set, we use the symbol $\notin$, e.g $7 \notin \{1,2,3,4\}$.

1.1.1 Set-builder Notations and Definitions

Sets are named using capital letters, while we use small letters for elements of the set. Capital letter such as A, B, C and so on, will be used to represent a set while lower case letters will be used to represent elements of a set. i.e. $\{a, b, c, d\}$ and so on. Numbers may also be used to represent the elements of a set.

A set can either be represented by writing/ listing down the elements, or by describing it. i.e. describing its elements. Each element in a set is separated from the other by a comma.

Definition 1.1.2 (Set-builder Notation). Sets are often written in set builder notation, which uses a variable such as x to describe the elements of the set. Example, describe the set $A = \{3, 4, 5, 6\}$ in set bulider notation.

solution $A = \{x \mid x \text{ is a natural number between 2 and 7}\}$, which is read “the set of all elements x such that x is a natural number between 2 and 7 . ” The symbol $\mid$ is sometimes replaced by $:$.

Consider the following examples:

<i>List</i>	<i>Description/Set-builder notation</i>
$A = \{1, 2, 3, 4, 5\}$	$A = \{x : x \text{ is a natural number less than } 6\}.$
$V = \{a, e, i, o, u\}$	$V = \{x : x \text{ is a vowel}\}.$
$C = \{2, 4, 6, 8, 10, \dots\}$	$C = \{x : x \text{ is a positive even number}\}.$
$E = \{\text{dog, plate, Earth, food}\}$	$E = \{\text{impossible}\}.$

We observe that it is possible to list the elements of the set from the given description.

Example 1.1.2. List down the elements of each of the following sets.

1. $A = \{x : x \text{ is an odd number}\}.$

$$2. B = \{k \in S : k = 3n + 1, n = 0, 1, 2, 3\}.$$

Solution

1.

$$A = \{1, 3, 5, 7, 9, \dots\}.$$

2.

$$B = \{1, 4, 7, 10\}.$$

Definition 1.1.3. Finite set

A set is said to be finite if its elements can be listed down. For instance, $V = \{x : x \text{ is a vowel}\}$ is a finite set.

Definition 1.1.4. infinite set

A set is said to be infinite if its elements can not be listed down in a finite period of time. For instance, $C = \{2, 4, 6, 8, 10, \dots\}$ is an infinite set, where the last three dots (points) show that the set continues according to the established pattern.

Definition 1.1.5. The *Universal set* usually denoted U/E is the set containing all the elements included in the discussion of a particular situation or problem.

Definition 1.1.6. The *null set or empty set* is the set containing no elements and normally denoted by $\emptyset$ or $\{\}$

Remark 1.1.2. $\{\emptyset\}$ is not an empty set. It is a set containing the symbol $\emptyset$.

Definition 1.1.7. Subset

A set S is a subset of T if every element of S is also an element of the set T , and we write, $S \subseteq T$, where the symbol $\subseteq$ denotes subset. If T has elements that are not elements of S , then we say that S is a proper subset of T and write $S \subset T$.

Example 1.1.3. 1. If $A = \{a, b, c\}$ and $B = \{a, c, b, d\}$, then $A \subset B$. Moreover, A is a proper subset of B .

2. Let $A = \{2, 5, 9\}$ and

$B = \{2, 3, 5, 6, 9, 10\}$. Then $A \subset B$.

Definition 1.1.8. Equal sets

Two sets A and B are said to be equal ($A = B$) if set A has the same members as set B regardless of the order of listing. In other words, two sets A and B are said to be equal if $A \subseteq B$ and $B \subseteq A$.

Example 1.1.4. If $A = \{a, b, c\}$ and $B = \{a, c, b\}$, then $A = B$.

Remark 1.1.3. 1. If a set A is not a subset of a set B , we write $A \not\subseteq B$, where the symbol $\not\subseteq$ means not a subset.

2. Every set is a subset of itself i.e, $A \subseteq A$. Where A is any set.

3. Empty set $\emptyset$ is a subset of every set, i.e $\emptyset \subseteq A$ where A is any set.

4. If a set A is of size or cardinality (the number of elements in a given set) n , then the total number of subsets of A is given by 2^n . The set consisting of all the subsets of the set A , denoted by $P(A)$, is called the *power set* of A .

Example 1.1.5. Let $A = \{1, 2, 3\}$, then A has cardinality 3 and so the total number of subsets of A is $2^3 = 8$. These subsets of A are $\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$.

1.2 Operations on sets

Definition 1.2.1. Complement of a set

The complement of a set A , denoted by A^c or A' , with respect to the given universal set U is the set of all elements of U that do not belong to A . That is

$$A^c = \{x : x \notin A, x \in U\}.$$

Example 1.2.1. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$. List down the elements of A^c .

Solution From definition 1.2.1,

$$A^c = \{2, 4, 6, 8, 10\}.$$

Definition 1.2.2. Union of sets

The union of two sets A and B , denoted by $A \cup B$, is the set of all elements that belong to A or B or both. That is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

Note that *definition 1.2.2* can be extended to more than two sets.

Example 1.2.2. Let $A = \{2, 3, 4\}$ and $B = \{4, 5, 6\}$. Find $A \cup B$.

Solution Clearly, by definition 1.2.2

$$A \cup B = \{2, 3, 4, 5, 6\}.$$

Recall that repetition of elements in a set is not allowed.

Definition 1.2.3. Intersection of sets

The intersection of two sets A and B , denoted by $A \cap B$, is the set of all elements that belong to both A and B . That is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Note that *definition 1.2.3* can be extended to more than two sets.

If for any two sets A and B , $A \cap B = \emptyset$, then A and B are said to be *disjoint*.

Example 1.2.3. Let $A = \{2, 3, 4\}$ and $B = \{4, 5, 6\}$. Find $A \cap B$.

Solution From definition 1.2.3, we get

$$A \cap B = \{4\}.$$

If A and B are sets, we write $A - B$, read as A minus B to mean the set of all elements which belong to A but do not belong to B . That is;

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

Thus, $A - B = A \cap B^c$.

Example 1.2.4. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 3, 4, 5\}$ and $B = \{1, 3, 4, 6\}$. List down the elements of $A - B$.

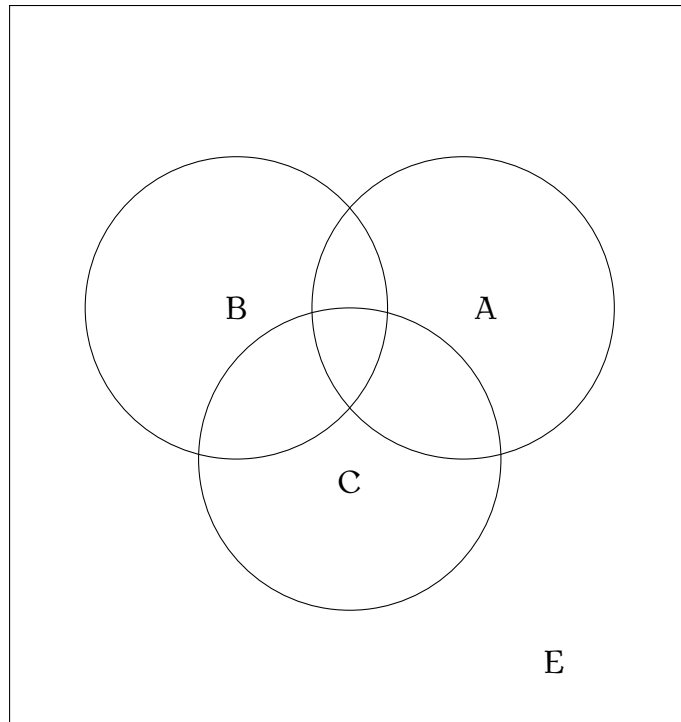
Solution

$$A - B = A \cap B^c = \{2, 5\}.$$

1.2.1 Venn diagram

A venn diagram is a pictorial representation of sets.

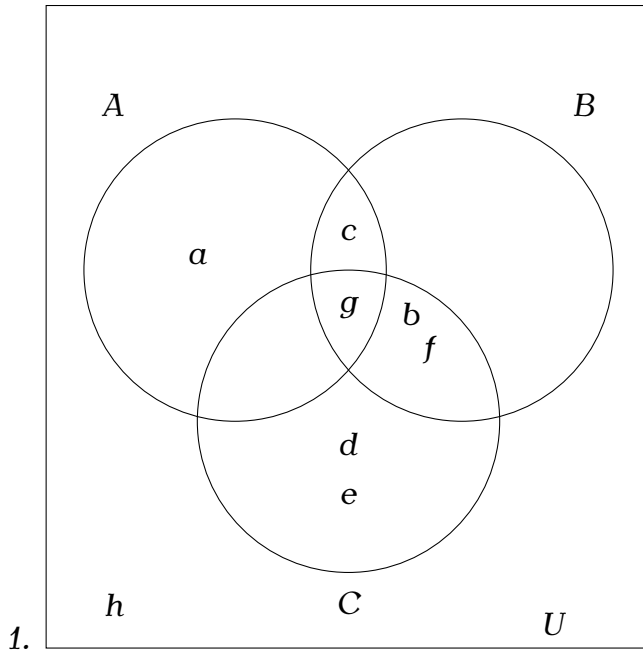
Union and intersection of sets may be described by using a Venn diagram. All the elements of a set are enclosed in a circle. For any three sets A , B and C , the union of the sets is the total area bounded by the circles in the Venn diagram, while the intersection is the overlapping cross-hatched area as shown below.



Example 1.2.5. Let $U = \{a, b, c, d, e, f, g, h\}$, $A = \{a, c, g\}$, $B = \{b, c, f, g\}$, $C = \{b, d, e, f, g\}$.

1. Illustrate this information on a Venn diagram.
2. Find $B \cup C$
3. Find A^c
4. Find $A^c \cap (B \cup C)$.

Solution



2. From the Venn diagram, it is easy to see that

$$B \cup C = \{b, c, d, e, f, g\}.$$

3. Clearly,

$$A^c = \{b, d, e, f, h\}$$

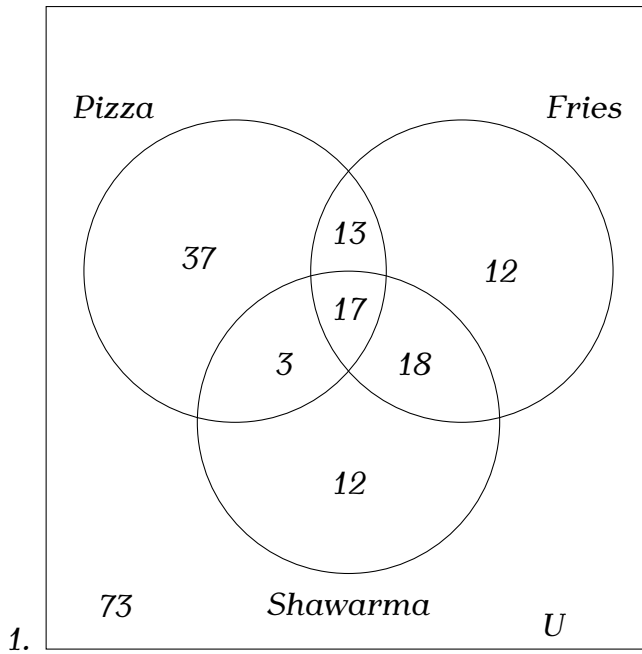
4. It follows from (2) and (3) that

$$A^c \cap (B \cup C) = \{b, d, e, f\}.$$

Example 1.2.6. A fast food restaurant conducted a survey on its 185 customers. The results came out as follows; 70 liked pizza, 60 liked fries, and 50 liked shawarma. In addition, 30 liked pizza and fries, 20 liked pizza and shawarma, 35 liked fries and shawarma. 17 liked all the three meals.

1. Illustrate this information on a Venn diagram.
2. Find the number of customers who did not like any of these available meals?
3. Find the number of customers who liked exactly one meal?

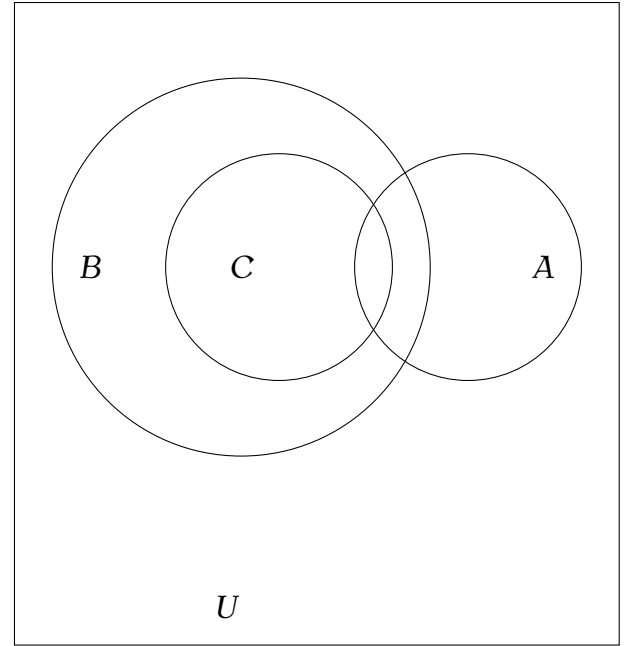
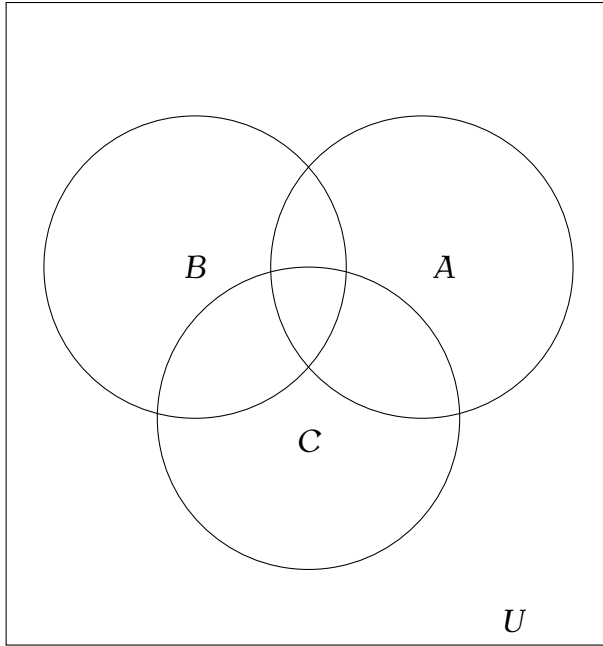
Solution



2. 73 customers did not like any of the three meals.
3. $37 + 12 + 12 = 61$ customers liked exactly one of the three meals.

Example 1.2.7. In each of the following Venn diagrams, shade

1. $B - C$.
2. $A \cap (B \cap C^c)$.



The following results will be very useful in performing certain set operations.

Theorem 1.2.1. *De Morgan's Laws of intersection ($\cap$) and union ($\cup$)*

Let A and B be any two sets. We have that

1.

$$(A \cap B)^c = A^c \cup B^c.$$

2.

$$(A \cup B)^c = A^c \cap B^c.$$

Example 1.2.8. *Consider the universal set*

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\},$$

and the subsets $A = \{2, 4, 6, 8\}$ and $B = \{3, 4, 5, 6\}$ of U . Confirm the above laws; the right hand side and left hand side are equal.

1.2.2 Laws of algebra of sets

Let A, B and C be any subsets of the universal set.

1. The identity law:

(a) $A \cup A = A$

(c) $A \cap \emptyset = \emptyset$

(e) $A \cup \emptyset = A$

(b) $A \cap A = A$

(d) $A \cap U = A$

(f) $A \cup U = U$.

2. Commutative law:

(a) $A \cup B = B \cup A$.

(b) $A \cap B = B \cap A$.

3. Associative law:

(a) $(A \cup B) \cup C = A \cup (B \cup C)$.

(b) $(A \cap B) \cap C = A \cap (B \cap C)$.

4. Distributive law:

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

5. Complement law:

(a) $A \cup A^c = U$

(b) $A \cap A^c = \emptyset$.

6. Complement law:

(a) $(A^c)^c = A$.

(c) $\emptyset^c = U$.

(e) $A \cap A^c = \emptyset$.

(b) $U^c = \emptyset$.

(d) $A \cup A^c = U$

Example 1.2.9. Given that $A \subset B$, simplify each of the following:

1.

$$[A^c \cup (A \cap B^c)]^c$$

2.

$$B^c \cap (A \cup B).$$

3.

$$A \cap (A \cup B).$$

Solution 1.2.1.

Chapter 2

Sets Of Numbers

2.1 Natural numbers

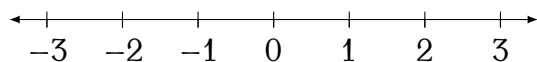
The first numbers we ever encounter in life are what are known as counting number, i.e, $1, 2, 3, 4, \dots$. This set of numbers is called the Natural numbers denoted by $\mathbb{N}$ and given by $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.

2.1.1 Whole numbers

If we include 0 to the set of natural numbers we have the set of whole numbers denoted by $\mathbb{W}$ and given by $\mathbb{W} = \{0, 1, 2, 3, 4, \dots\}$.

2.1.2 Integers

Including the negatives of natural numbers to set of whole numbers gives the set of integers denoted $\mathbb{Z}$ and given by $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. Integers can be represented on a number line as shown in the figure below;



Note It is important so far to note that $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z}$.

If one divides two integers with a non-zero divisor, one obtains a fraction also known as a rational number.

Definition 2.1.1. A *rational number* is an element of the set defined as follows:

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \text{ and } q \text{ are integers and } q \neq 0 \right\} \quad (2.1.1)$$

Remark 2.1.1. The set of rational numbers $\mathbb{Q}$ includes the natural numbers, the whole numbers and integers. E.g -3 is a rational number since it can be written as $\frac{-3}{1}$. Rational numbers are numbers that can be written as,

1. Terminating decimals e.g $\frac{3}{4} = 0.75$
2. Repeating decimals e.g $\frac{23}{11} = 2.090909\ldots = 2.\overline{09}$ where the bar indicates the block of repeating digits.

Again, we note so far that $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q}$.

Example 2.1.1. Show that each of the following numbers are rational.

1. $25.\overline{12}$

2. $0.3\overline{59}$.

Solutions

1.

Let $x = 25.\overline{12}$. Multiplying by 100 both sides yields.

$100x = 2512.\overline{12}$. Thus subtracting equation 2 from equation 1 yields,

$$100x - x = 2512.\overline{12} - 25.\overline{12}.$$

$$\Rightarrow 99x = 2487,$$

$$\Rightarrow x = \frac{2487}{99} = \frac{829}{33}$$

2.

Let $x = 0.\overline{359}$. Multiplying by 10 both sides yields.

$10x = 3.\overline{59}$. Multiplying both sides by 100 yields,

$1000x = 359.\overline{59}$. Thus subtracting equation 3 from equation 2 yields,

$$1000x - 10x = 359.\overline{59} - 3.\overline{59}.$$

$$\Rightarrow 990x = 356,$$

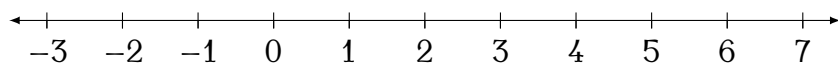
$$\Rightarrow x = \frac{356}{990} = \frac{178}{495}.$$

2.1.3 Real Numbers

The set of all numbers that correspond to any point on the number line is the set of real numbers. This set is denoted by $\mathbb{R}$. Real numbers are numbers that can be expressed as terminating decimals or non-terminating decimals. These numbers includes

$$4.00\ldots, \frac{-3}{4} = -0.7500\ldots, \frac{1}{3} = 0.3333\ldots, \text{ and } \sqrt{2} = 1.4142\ldots.$$

The dots ($\ldots$) in each case indicate that the sequence of decimal digits goes on forever. Real numbers can be represented geometrically as points on a (straight) number line called the real line as shown below.



Remark 2.1.2. 1. We note that some real numbers can be represented by terminating decimals, e.g $\frac{1}{4} = 0.25$ is a real number, hence real numbers include rational numbers.

2. Some real numbers cannot be represented by quotients / fractions of integers. These numbers are called **Irrational numbers** denoted by $\mathbb{Irr}$. The set of irrational numbers include $\pi, \sqrt{2}, \sqrt{5}, \sqrt{2} + \sqrt{3}$. Thus irrational numbers are real numbers that are not rational.

3. $(a, b]$ represents all the real numbers between a and b , not including a but including b . This is a half open interval. In set builder notation, we write $\{x : a < x \leq b, x \in \mathbb{R}\}$. For example, the graph of $(-1, 3]$ is

4. $[a, b)$ represents all the real numbers between a and b , including a but not b . This is a half open interval. In set builder notation, we write $\{x : a \leq x < b, x \in \mathbb{R}\}$. For example, the graph of $[-4, -1)$ is

Subsets of the real numbers whose graphs extend forever in one or both directions can be represented by interval notation using the infinity symbol ∞ or the negative infinity symbol $-\infty$. These are:

1. $(-\infty, a)$
2. (b, ∞)
3. $(-\infty, a]$
4. $[b, \infty)$
5. $(-\infty, \infty)$

Example 2.1.2. Let $U = [0, 10)$ and define $A = [0, 5]$, $B = [2, 8]$ and $C = (1, 6)$. Find each of the following sets and display them on the number line.

1. $A \cap C$

3. $A \cap B^c$

2. $A \cap B$

4. $A^c \cap C^c$

Solution 2.1.1. *We Solve as follows*

2.2 Radicals and rational powers

In this section, we will learn how to simplify expressions involving square roots, rationalize denominators and use rational powers/indices.

2.2.1 Square roots

If $b^2 = a$, then b is called a square root of a . The symbol $\sqrt{}$ is used to denote the positive or principal square root of a number, while $-\sqrt{}$ denotes the negative square root.

1. $\sqrt{25} = 5$ because $5^2 = 25$ and 5 is positive.

2. $\sqrt{100} = 10$ because $10^2 = 100$ and 10 is positive.

The symbol $\sqrt{}$ that we use to denote the positive square root is called a *Radical sign*. The number under the radical sign is called the *Radicand*. Together $\sqrt{a}$ is called a *radical expression*.

Example 2.2.1. *Evaluate*

1. $\sqrt{64}$

2. $\sqrt{9} + \sqrt{16}$

A rational number that is a square of another rational number is called a *perfect square*. All the radicands in the previous example are perfect squares. $\frac{1}{4}$ is a perfect square because $(\frac{1}{2})^2 = \frac{1}{4}$.

Theorem 2.2.1. *Product rule of square roots*

If a and b are positive real numbers, then

$$\sqrt{ab} = \sqrt{a}\sqrt{b}.$$

Example 2.2.2. *Use the product rule to simplify*

1. $\sqrt{500}$

2. $\sqrt{6x} \times \sqrt{3x}$

Another property for square roots involves division.

Theorem 2.2.2. *Quotient rule of square roots*

If a and b are positive real numbers, then

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

Example 2.2.3. *Use the quotient rule to simplify*

1. $\sqrt{\frac{100}{9}}$

2. $\frac{\sqrt{48x^3}}{\sqrt{6x}}$

Adding and subtracting square roots

Two or more square roots can be combined using the distributive property provided that they have the same radicands. Such radical expressions are called like radicals. For example, $7\sqrt{11} + 6\sqrt{11} = (7 + 6)\sqrt{11} = 13\sqrt{11}$.

Example 2.2.4. *Add or subtract as indicated.*

1. $7\sqrt{2} + 5\sqrt{2}$

3. $7\sqrt{3} + \sqrt{12}$

2. $\sqrt{5x} - 7\sqrt{5x}$

4. $4\sqrt{50x} - 6\sqrt{32x}$

Solution 2.2.1. 1. $7\sqrt{2} + 5\sqrt{2}$

2.2.2 Rationalizing the denominator

Rationalizing the denominator is the process that involves rewriting a radical expression as an equivalent expression in which the denominator no longer contains any radical sign.

If the denominator consists of the square root of a natural number that is not a perfect square, multiply the numerator and the denominator by the smallest

radical expression that produces the square root of a perfect square in the denominator.

Example 2.2.5. *Rationalize the denominator.*

1. $\frac{15}{\sqrt{6}}$

2. $\frac{12}{\sqrt{8}}$

Solution 2.2.2. 1. $\frac{15}{\sqrt{6}} =$

Radical expressions that involve the sum and difference of the same two terms are called *conjugates*. For example, $\sqrt{a} + \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ are conjugates of each other. What happens when two conjugates are multiplied? Let us see

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$$

To rationalize the denominator if the denominator contains two terms with one or two square roots, we multiply the numerator and the denominator by the conjugate of the denominator.

Example 2.2.6. *Rationalize the denominator.*

$$1. \frac{7}{\sqrt{5}+\sqrt{3}}$$

$$2. \frac{h}{\sqrt{x+h}-\sqrt{x}}$$

Solution 2.2.3. 1. $\frac{7}{\sqrt{5}+\sqrt{3}} =$

Example 2.2.7. *Simplify*

$$1. \left(\frac{\sqrt{3}}{21}\right)^{-1} + \sqrt{27}.$$

$$2. -1 + \sqrt{125} + \frac{1}{1-\sqrt{5}}.$$

Solution 2.2.4. 1. $\left(\frac{\sqrt{3}}{21}\right)^{-1} + \sqrt{27} =$

Similarly, one can easily rationalize the numerator by considering the above discussion.

2.3 Complex numbers

A complex number is an expression of the form $a + bi$, where $a, b \in \mathbb{R}$, and $i^2 = -1$ or equivalently $\sqrt{-1} = i$. The set of such numbers is called the set of complex numbers and is denoted by $\mathbb{C}$.

If $z = a + bi \in \mathbb{C}$, then we call a the real part of z and b the imaginary part of z . We write this as $a = \operatorname{Re}(z)$ and $b = \operatorname{Im}(z)$. The representation of a complex number in the form $a + bi$ is called the Cartesian form of the complex number. If $b = 0$, then $z = a$, and z is said to be purely real. Similarly, z is purely imaginary if $a = 0$.

Example 2.3.1. Let $z = -3 - 4i$. State $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$.

2.3.1 Equal complex numbers

Definition 2.3.1. Two complex numbers $z_1 = a + bi$ and $z_2 = c + di$ are equal ($z_1 = z_2$) if and only if $a = c$ and $b = d$.

Example 2.3.2. 1. State the value of x and of y given that $x + yi = 5 + 4i$.

2. Find the value of a and b if $(a + b) + (a - b)i = 7 + 2i$.

Solution 2.3.1. 1. $x = 5$ and $y = 4$.

2.

2.3.2 Addition and subtraction of complex numbers

Let $z = a + bi$ and $\omega = c + di$ be two complex numbers.

$$z + \omega = (a + bi) + (c + di).$$

Although the real and the imaginary parts can not be combined, we can remove the brackets and total up like terms. That is

$$\begin{aligned} z + \omega &= a + bi + c + di \\ &= (a + c) + bi + di \\ &= (a + c) + (b + d)i. \end{aligned}$$

Example 2.3.3. Compute $z + \omega$ if $z = 4 + 5i$ and $\omega = 3 - 2i$.

Similarly, if $z = a + bi$ and $\omega = c + di$, then

$$\begin{aligned} z - \omega &= (a + bi) - (c + di) \\ &= a + bi - c - di \\ &= (a - c) + (b - d)i. \end{aligned}$$

Example 2.3.4. Find $z - \omega$ if $z = 4 + 5i$ and $\omega = 3 - 2i$.

2.3.3 Multiplication of complex numbers

Let $z = a + bi$ and $\omega = c + di$ be two complex numbers.

$$z\omega = (a + bi)(c + di).$$

These are multiplied in the same way as you would determine the product $(3x + 4y)(2x + 5y)$. Thus,

$$\begin{aligned} z\omega &= (a + bi)(c + di) \\ &= ac + adi + bi(c) + bdi \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i. \end{aligned}$$

Example 2.3.5. Given that $z = 3 + 4i$ and $\omega = 2 + 5i$. Compute $z\omega$.

Solution 2.3.2. $z\omega =$

If the expression contains more than two factors, we multiply the factors together in stages.

Example 2.3.6. Multiply $(3 + 4i)(5 - 8i)(1 - 2i)$.

From the previous example, we observed that $i^3 = -i$, because $i^2 = -1$. Continuing, we see that $i^4 = i^2 \times i^2 = -1 \times -1 = 1$, $i^5 = i$ and $i^6 = -1$.

In summary, to calculate any high power of i , you can convert it to a lower power by taking the closest multiple of 4 that is no longer bigger than the exponent and subtract this from the exponent. For example, $i^{99} = i^{(96+3)} = i^3 = -i$.

Example 2.3.7. *Simplify each of the following:*

1. i^{17}

2. i^{120}

3. i^{64002}

2.3.4 Conjugate complex numbers

Compute $(5 + 8i)(5 - 8i)$.

A pair of complex numbers like these are called *conjugate complex* numbers, and the product of conjugate complex numbers is always purely real. In general if $z = a + bi$, then the conjugate of z , denoted by $\bar{z}$, is given by $\bar{z} = a - bi$.

Example 2.3.8. Given that $z = 5 + 4i$ compute

1. $z - \bar{z}$

2. $z + \bar{z}$

3. $z\bar{z}$

Solution 2.3.3. 1. $z - \bar{z} =$

2.3.5 Division of complex numbers

Division of a complex number by a real number is very easy. For example, if $z = 5 - 4i$, then $\frac{z}{3} = \frac{5-4i}{3} = \frac{5}{3} - \frac{4}{3}i$. However, how do we manage with $\frac{7-4i}{4+3i}$?

If we could convert the denominator into a real number, we could divide out as in the example above. This can easily be done by multiplying the denominator by its complex conjugate.

Example 2.3.9. Simplify each of the following:

1.

$$\frac{3 + 2i}{1 - 3i}$$

2.

$$\frac{1 + i}{3 - 2i} + 2 + 4i$$

Solution 2.3.4.

1.

$$\frac{3 + 2i}{1 - 3i} =$$

Example 2.3.10. Solve each of the following for a and b .

1.

$$\frac{3i - 2}{2i - 6} = 3a + 3(b - 1)i.$$

2.

$$\frac{2}{i} - 2b + i^3 = \frac{i + a}{i} + b.$$

Solution 2.3.5.

1.

$$\frac{3i - 2}{2i - 6} = 3a + 3(b - 1)i =$$

